

DBMS Normalization

What is Normalization?

Normalization is the process of organizing data in a database to:

- Reduce data redundancy
- Avoid data inconsistency
- Improve data integrity
- Organize tables properly

Normalization divides large tables into smaller related tables.

Why Normalization is Needed?

Without normalization:

- Duplicate data increases
- Storage waste happens
- Updating becomes difficult
- Data inconsistency occurs

Example:

StudentID	StudentName	Course	Teacher
-----------	-------------	--------	---------

1	Rahul	DBMS	Sharma
---	-------	------	--------

1	Rahul	OS	Verma
---	-------	----	-------

Here Rahul's name repeats multiple times.

Types of Anomalies

1. Insertion Anomaly

Cannot insert data properly without other data.

Example:

Cannot add a new course until a student enrolls.

2. Deletion Anomaly

Deleting one record may remove important information.

Example:

Deleting last student removes course information too.

3. Update Anomaly

Same data repeated in many places must be updated everywhere.

Example:

Teacher name change requires updating multiple rows.

Functional Dependency (FD)

A functional dependency means:

$A \rightarrow B$

Meaning:

If A is known, B can be determined.

Example:

StudentID $\rightarrow$ StudentName

StudentID uniquely determines StudentName.

Types of Keys

1. Super Key

Set of attributes that uniquely identifies a row.

Example:

(StudentID)

(StudentID, Name)

2. Candidate Key

Minimal super key.

Example:

StudentID

3. Primary Key

Chosen candidate key.

Example:

StudentID

4. Foreign Key

Key used to connect two tables.

Normal Forms

First Normal Form (1NF)

Rule

- Each column must contain atomic values
- No multi-valued attributes

Not in 1NF

Student Phones

Rahul 123,456

Convert to 1NF

Student Phone

Rahul 123

Rahul 456

Second Normal Form (2NF)

Rule

- Table must already be in 1NF
- No partial dependency

Partial dependency occurs when non-prime attribute depends on part of composite key.

Example

StudentID CourseID StudentName CourseName

Composite Key:

(StudentID, CourseID)

Problem:

$\text{StudentID} \rightarrow \text{StudentName}$

$\text{CourseID} \rightarrow \text{CourseName}$

Partial dependency exists.

Convert to 2NF

Student Table

| StudentID | StudentName |

Course Table

| CourseID | CourseName |

Enrollment Table

| StudentID | CourseID |

Third Normal Form (3NF)

Rule

- Table must be in 2NF
 - No transitive dependency
-

Transitive Dependency

$A \rightarrow B$

$B \rightarrow C$

Then:

$A \rightarrow C$

Indirect dependency exists.

Example

| EmpID | EmpName | DeptID | DeptName |

Dependencies:

$\text{EmpID} \rightarrow \text{DeptID}$

$\text{DeptID} \rightarrow \text{DeptName}$

Transitive dependency exists.

Convert to 3NF

Employee Table

| EmpID | EmpName | DeptID |

Department Table

| DeptID | DeptName |

Boyce-Codd Normal Form (BCNF)

Rule

For every dependency:

$X \rightarrow Y$

X must be a super key.

BCNF is stronger than 3NF.

Example

| Student | Subject | Teacher |

Dependency:

Subject $\rightarrow$ Teacher

If Subject is not super key $\rightarrow$ violates BCNF.

Fourth Normal Form (4NF)

Rule

- Must be in BCNF
 - No multivalued dependency
-

Example

| Student | Hobby | Language |

A student can have multiple hobbies and languages independently.

Separate tables are created.

Fifth Normal Form (5NF)

Rule

- Removes join dependency

- Decomposes tables further without data loss

Used rarely.

Normalization Summary Table

Normal Form Removes

1NF	Repeating groups
2NF	Partial dependency
3NF	Transitive dependency
BCNF	Non-super key dependency
4NF	Multivalued dependency
5NF	Join dependency

Advantages of Normalization

- Reduces redundancy
- Improves consistency
- Easier maintenance
- Better integrity
- Efficient updates

Disadvantages of Normalization

- More joins required
- Query complexity increases
- Can reduce performance in some cases

Denormalization

Opposite of normalization.

Combines tables to improve read performance.

Used in:

- Data Warehouses
- Analytics systems

Most Important Interview Questions

Q1. What is normalization?

Process of organizing data to reduce redundancy and anomalies.

Q2. Difference between 2NF and 3NF?

- 2NF removes partial dependency
 - 3NF removes transitive dependency
-

Q3. What are anomalies?

- Insertion anomaly
 - Deletion anomaly
 - Update anomaly
-

Q4. What is BCNF?

Every determinant must be a candidate/super key.

Q5. Difference between BCNF and 3NF?

3NF may allow some non-super key dependencies.
BCNF removes them completely.

Important Functional Dependency Formulas

Partial Dependency

$(A,B) \rightarrow C$

$A \rightarrow C$

Transitive Dependency

$A \rightarrow B$

$B \rightarrow C$